

$\therefore 3$ is the counterexample to the statement that every positive integer is the sum of the squares of two integers.

1.7.8 Mistakes in proofs

Example 16: What is wrong with this famous supposed "proof" that $1=2$?

Proof: We use these steps, where a and b are two equal positive integers.

<u>Step</u>	<u>Reason</u>
1. $a=b$	Given
2. $a^2=ab$	Multiply both sides of (1) by a
3. $a^2-b^2=ab-b^2$	Subtract b^2 from both sides of (2)
4. $(a-b)(a+b)=b(a-b)$	Factor ^{both} sides of (3)
5. $a+b=b$	Divide both sides of (4) by $a-b$
6. $2b=b$	Replace a by b in (5) because $a=b$ and simplify
7. $2=1$	Divide both sides of (6) by b .

Solution: The error is in Step 5. Since in (1) we know $a=b$, $\therefore a-b=0$. We cannot divide by 0 since it's an undefined operation. $\therefore$ We cannot proceed from Step 4 to Step 5.

Many incorrect arguments are based on a fallacy called begging the question. This fallacy occurs when one or more steps of a proof are based on the truth of the statement being proved. In other words, this fallacy arises when a statement is proved using itself, or a statement equivalent to it. That is why this fallacy is also called circular reasoning.